

A Review of Machine Learning, Deep Learning and Ensemble Learning Approaches to Celestial Entity Morphological Analysis

With a Particular Focus on Multi-Channel Inputs

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1) Abstract Classifying celestial objects involves a deep understanding of the type of data being used, as it varies significantly depending on the wavelength of observation. Data collected below the visual spectrum (such as radio or infrared waves) reveals different characteristics of celestial objects compared to data in the visual spectrum, which provides a more familiar view of galaxies and stars. Similarly, data collected above the visual spectrum, such as X-rays or gamma rays, offer insight into high-energy processes and phenomena. Understanding how to interpret these different types of telemetry is crucial for accurate classification and analysis.

(key words) - Morphology, Machine vision, Galaxy classification, Image interpretation

II. INTRODUCTION

As astronomical observation technology progresses and the construction of telescopes proliferates, the volume of collected data has expanded significantly. This growth presents a substantial challenge in the timely processing and classification of data. Current data analysis workflows often struggle to keep pace with the influx of new information, resulting in delays that can leave some data from telescopic surveys unexamined for weeks or even months. This lag can postpone scientific insights and discoveries. Moreover, much of this data is presented in telemetry format without visual representation, meaning that detailed images of observed celestial structures are not readily accessible. A machine learning-based classification model could address these issues by expediting the classification process, enabling researchers to focus on more complex tasks such as anomaly detection and in-depth analysis, rather than repetitive categorization. For instance, papers such as “Galaxy Spectral Classification and Feature Analysis Based on Convolutional Neural Networks” [1] work with 1D spectral data to classify galaxies based on emissions, while “Discovery and Classification in Astronomy: Scientific and Philosophical Challenges and the Importance of a Comprehensive and Consistent Classification System” [2] highlights the metaphysical questions faced by researchers in classifying celestial structures. While many efforts in astronomical classification tackle specific tasks, this research aims to develop an analysis tool that recognizes the visual characteristics of galaxies. The research will emphasize a logistic process that can be expanded on for more robust results.

III. ASTRONOMY

A. Limiting Factors

Astronomy is a field characterized by its vastness and complexity, yet it faces several significant challenges. One of the main obstacles is the limited number of researchers involved, particularly given the immense amount of data being generated. While the field itself holds immense potential for discovery, there are few financial incentives driving individuals or organizations to explore the depths of the universe. As a result, resources—both human and financial—are often stretched thin, leaving many data sets under-explored or delayed in processing. Coupled with the growing volume of data from telescopes and space missions, this imbalance between the available resources and the sheer size of the task becomes increasingly difficult to manage.

B. Data Interpability

The majority of high-quality astronomical data is gathered through various observation methods, which often generate raw telemetry data. While these data sets contain vital information about celestial objects, they are not immediately accessible in a form that researchers can easily and intuitively interpret. To make sense of this raw data, it must be processed into visual images, or other means that allow for the detailed analysis of galaxies, stars, and other cosmic phenomena. Machine learning is particularly well-suited for this task, as it can efficiently process vast amounts of telemetry data, far quicker and more accurately than humans can. This capability enables faster classification of celestial objects and the identification of patterns that might otherwise go unnoticed.

C. Scientific Capital

One of the challenges in astronomy is that even when high-quality data is available, it is not always aligned with the specific research goals of astronomers, making it difficult to consistently identify intriguing objects. Missions like Galaxy Zoo address this issue by focusing on large-scale observations that help catalog and classify objects across vast regions of space, ensuring that valuable data does not go unnoticed.

IV. OUTLINING THE PROBLEM

Despite the increasing number of astronomical facilities and advanced telescopes being launched worldwide, there remains a critical shortage of manpower to analyze the vast amounts of data they produce. As more observatories come online and collect high-quality data, the challenge of sifting

through it to identify the most significant objects or phenomena grows. Analyzing data across different wavelengths—such as radio, infrared, visual, and X-ray—requires specialized expertise in each spectral range, as each provides distinct information about celestial objects. The sheer volume of data, coupled with the need for specific knowledge in each spectrum, far exceeds the capacity of the limited number of researchers available to analyze it. This results in delays and missed opportunities for discovery. The shortage of human resources makes it difficult to keep pace with the rapidly expanding field of astronomy, underscoring the need for automated tools to help process and interpret the data more efficiently.

V. DATA DESCRIPTION

A. About GalaxyZoo

This study will utilize the galaxy datasets hosted by Galaxy Zoo. Galaxy Zoo is a citizen science project that invites the public to help classify galaxies based on their shapes. By analyzing these shapes, volunteers help scientists study galaxy formation, evolution, and interactions. The project uses these classifications to train AI algorithms, enabling the exploration of large datasets from space telescopes like Hubble and Euclid, advancing our understanding of the Universe. Huge thanks to the project for making the dataset widely available and having so many resources to work with.

B. About the Data

The GZ2 dataset has 167434 entries, each containing image data and labels for the images. Each image data point contains three images representing different color channels (Red, Green, and Blue) with a resolution of 224 224 pixels. Each label data point contains 30 integer values representing a score from 0 - 80 for the following visual attributes: 'smooth-or-featured-gz2_smooth', 'smooth-or-featured-gz2_featured-or-disk', 'smooth-or-featured-gz2_artifact', 'disk-edge-on-gz2_yes', 'disk-edge-on-gz2_no', 'has-spiral-arms-gz2_yes', 'has-spiral-arms-gz2_no', 'bar-gz2_yes', 'bar-gz2_no', 'bulge-size-gz2_dominant', 'bulge-size-gz2_obvious', 'bulge-size-gz2_just-noticeable', 'bulge-size-gz2_no', 'something-odd-gz2_yes', 'something-odd-gz2_no', 'how-rounded-gz2_round', 'how-rounded-gz2_in-between', 'how-rounded-gz2_cigar', 'bulge-shape-gz2_round', 'bulge-shape-gz2_boxy', 'bulge-shape-gz2_no-bulge', 'spiral-winding-gz2_tight', 'spiral-winding-gz2_medium', 'spiral-winding-gz2_loose', 'spiral-arm-count-gz2_1', 'spiral-arm-count-gz2_2', 'spiral-arm-count-gz2_3', 'spiral-arm-count-gz2_4', 'spiral-arm-count-gz2_more-than-4', and 'spiral-arm-count-gz2_cant-tell'.



Figure 1: an example picture printed in different channels

C. Prior Works

There has been lots of research conducted in the field of morphological analysis and classification of celestial bodies.

The study "Combining Human and Machine Learning for Morphological Analysis of Galaxy Images" [3] explores how citizen science data from the Galaxy Zoo 2 project can be used to train machine learning models for galaxy image classification. It emphasizes that high-quality, consistent annotations from citizen scientists improve the performance of machine learning algorithms, which is a concept central to my work as well. Both the study and this research leverage machine learning for galaxy classification, but while the study uses traditional feature extraction methods like Wndchrm, this paper will review approaches that employ deep learning models, specifically convolutional neural networks (CNNs), which directly extract features from raw images. Like the study, this paper focuses on improving the quality of the training data by ensuring consistent annotations, utilizing techniques like data cleaning and setting agreement thresholds to filter out less reliable data.

Another closely aligned example would be "*Machine and Deep Learning Applied to Galaxy Morphology - A Comparative Study*" [4], which investigates galaxy morphology classification using both traditional machine learning (TML) and deep learning (DL) techniques. Like the study in this paper, this research employs Galaxy Zoo classifications as the ground truth and utilizes SDSS data to train the classification models. Both approaches aim to classify galaxies into different morphological types, using large datasets to overcome the challenges of manual classification. The paper compares the effectiveness of traditional machine learning algorithms, such as decision trees and random forests, with deep learning models, primarily convolutional neural networks (CNNs), in classifying galaxies as elliptical or spiral types. This research, however, integrates ensemble learning methods to combine multiple models for enhanced performance and uses CNNs to improve classification accuracy. Additionally, while the paper tackles issues such as class imbalance through different sampling techniques, this research also emphasizes dimensionality reduction methods like t-SNE and UMAP to better visualize and interpret galaxy data, and to highlight the impact of RGB channels on model performance. Furthermore, both studies use ensemble methods to boost classification accuracy—combining multiple annotations or models to create a more robust system. In essence, while the study combines citizen science with machine learning to classify galaxies, my work builds upon this by integrating deep learning models to handle larger datasets more effectively.

VI. DATA PREPROCESSING

In the process of preparing and analyzing image data for neural network models, several crucial steps are involved to ensure that the data is properly formatted, cleaned, and scaled for optimal model training. This process starts with a dataset containing 10,000 images, which are initially in a raw, unprocessed format, and concludes with a sampled subset of 5,000 images. Below is an expanded explanation of each key step in this data preparation process: image data extraction, label scaling, data reshaping, and cleaning.

A. Data Extraction and Preprocessing:

The first step in this process is the extraction of image data from the dataset. Typically, this is done using a TensorFlow dataset iterator. TensorFlow is a popular open-source machine learning library, and its dataset iterator allows for efficient management of large datasets by enabling the loading of data

in batches. In the context of image classification tasks, the iterator is used to read images and their corresponding labels, ensuring that data can be processed in manageable chunks during training. This extraction process typically starts with a large dataset, such as one containing 10,000 images, and ensures that each image and its label are correctly paired.

As part of the extraction process, the image data is separated into individual red, green, and blue (RGB) channels. Images are composed of three primary color channels: red, green, and blue. By separating the image into these channels, the data becomes more manageable for neural networks, which often process the individual channels separately to learn relevant patterns. This separation also ensures consistency across the dataset, as each channel is handled in a uniform way, ensuring that no single color channel dominates the learning process.

B. Label Scaling

Once the image data is extracted and the color channels are separated, the next step is to scale the label data. In many classification tasks, labels can be quite diverse or may span a large range of values. To simplify the learning process for the neural network, label scaling is used. In this case, the Min-Max Scaler is employed, which scales the labels from a range of 0-80 down to 0-8. This scaling step is crucial because neural networks generally perform better when the values of the labels are smaller and lie within a consistent range. The Min-Max scaling maps the original labels linearly to the new range, ensuring that the network can more easily adjust during training and makes predictions more efficient.

The purpose of label scaling is not just to normalize the values but also to ensure that the network doesn't become overwhelmed by excessively large label values, which could hinder learning. This scaling step provides the neural network with a more consistent range of target values, leading to a more effective and stable training process.

C. Data Reshaping and Cleaning

After scaling the labels, the next step is to reshape the image data for input into the neural network. Initially, the raw image data is in a multidimensional format (for example, a 3D array where the dimensions represent height, width, and color channels). For many neural networks, especially fully connected networks, the image data needs to be flattened into a 1D array. This transformation simplifies the data, making it easier for the network to process. Each image is flattened into a vector, where each pixel from each channel is mapped to a single number in the vector.

For example, a 32x32 image with three color channels would be reshaped into a single vector containing $32 \times 32 \times 3 = 3,072$ elements. This reshaping step allows the neural network to receive data in a format that it can process more easily, enabling better learning performance. Alongside reshaping, data cleaning is performed. This involves checking for empty or invalid entries in the dataset. For example, some images might have been corrupted or contain missing data, and it is essential to remove such entries to prevent them from affecting the training process. Cleaning ensures that only valid, complete images are used to train the neural network.



Figure 2: visual representation of reshaping image data.

D. Data Sampling

At this stage, a subset of the data is sampled for more detailed analysis. From the original dataset of 10,000 images, 5,000 images are selected for further examination. Sampling is a critical step in data analysis, as it allows for the creation of a more manageable dataset that can be used to assess the effectiveness of various preprocessing and modeling techniques. By choosing a smaller, representative subset, the model can be trained on a manageable amount of data without compromising the diversity or complexity of the entire dataset.

The selection of 5,000 images provides the model with a diverse set that facilitates insights into how well the neural network generalizes to the full dataset. This sampled data is then utilized to generate detailed reports on the image data, including visualizations of the data distribution and labels, as well as performance metrics of the models employed in the research. In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)

VII. DIMENSIONALITY REDUCTION AND VISUALIZATION

A. t-SNE

t-distributed Stochastic Neighbor Embedding (t-SNE) is a machine learning algorithm for dimensionality reduction, particularly well-suited for visualizing high-dimensional datasets. It maps multi-dimensional data to two or three dimensions for visualization, highlighting patterns like clusters or gradients. Since t-SNE is optimal for recognizing local relationships, it was used to visualize the potential relationship of local characteristics such as spiral arm count in reduced dimensions.

B. UMAP

Uniform Manifold Approximation and Projection (UMAP) is another dimensionality reduction technique that is highly effective for visualizing complex data structures. It preserves more of the global structure of the data compared to t-SNE, making it useful for understanding underlying patterns and relationships. Since UMAP is better suited for recognizing global relationships, it was used to visualize the potential relationship of global characteristics such as rounded in reduced dimensions.

VIII. MODELS

A. DNN

A Dense Neural Network (DNN), also known as a fully connected network, is one of the fundamental architectures in artificial neural networks. It consists of multiple layers of neurons where each neuron in one layer is connected to every neuron in the subsequent layer. These layers typically include an input layer, one or more hidden layers, and an output layer. Each neuron applies an activation function, such as ReLU (Rectified Linear Unit) or sigmoid, to introduce non-linearity into the network, enabling it to learn complex patterns in the data. DNNs are trained using a process called backpropagation, which adjusts the weights of the connections between neurons to minimize the error in predictions. This architecture is versatile and widely used in tasks such as classification, regression, and pattern recognition, effectively capturing intricate relationships within the data.

B. CNN

Convolutional Neural Network (CNN) is a type of deep learning model particularly well-suited for processing structured grid data, such as images. Unlike traditional neural networks, CNNs utilize convolutional layers to scan across the input data with filters (or kernels) that detect spatial hierarchies of features. These layers are followed by pooling layers, which down sample the data to reduce its dimensionality and highlight the most important features. Activation functions like ReLU introduce non-linearity, allowing the network to learn complex patterns. The final layers are typically fully connected (dense) layers that combine the features detected by the convolutional layers to make predictions. CNNs are widely used in tasks such as image classification, object detection, and facial recognition, due to their ability to automatically and efficiently learn intricate features from raw pixel data.

C. Ensemble Learner

Ensemble Learner is a machine learning technique that combines the predictions from multiple models to improve overall performance and robustness. The idea is to leverage the strengths of diverse models and mitigate their individual weaknesses, resulting in better generalization to new data. There are several methods to create ensembles, including Bagging, Boosting, and Stacking.

This paper focuses on stacking three models (one for each color channel) and having the meta learner sum up their voting in regard to the target classification. This method is interesting since the different learners have an opportunity to fit color specific structures. An interesting follow up would be to implement the same technique using data from different wavelength like an X-ray learner combined with a radio learner.

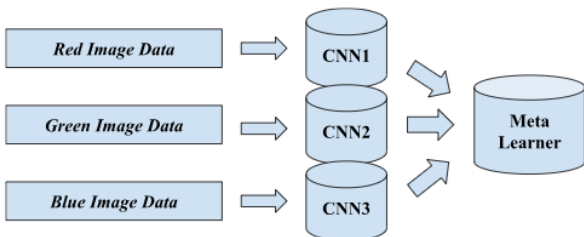


Figure 2: ensemble model visualization

IX. RESULTS

A. Data Visualization

t-SNE analysis was conducted in 2D on the red channel images since light experiences "red-shifting" which would leave the most information even for farther objects. No noticeable structures are present in most of these except for a few.

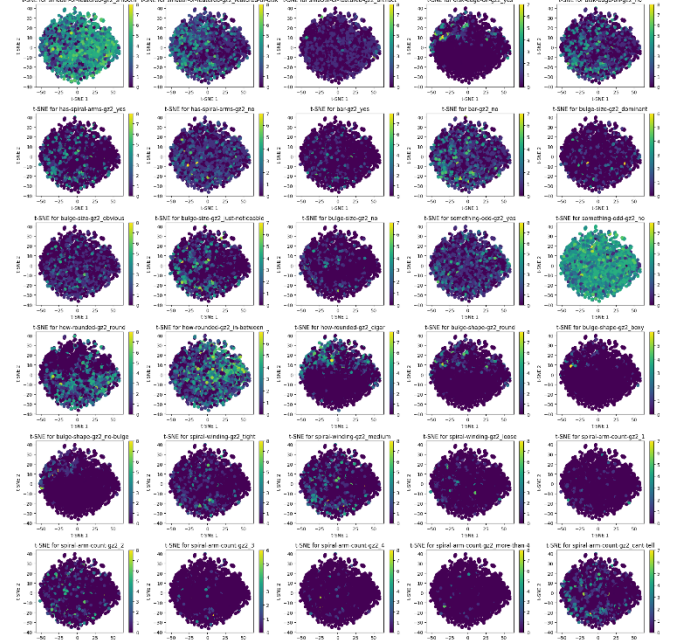


Figure 4: t-SNE analysis conducted for each target label

UMAP analysis was conducted in 2D on red channel image data as well.

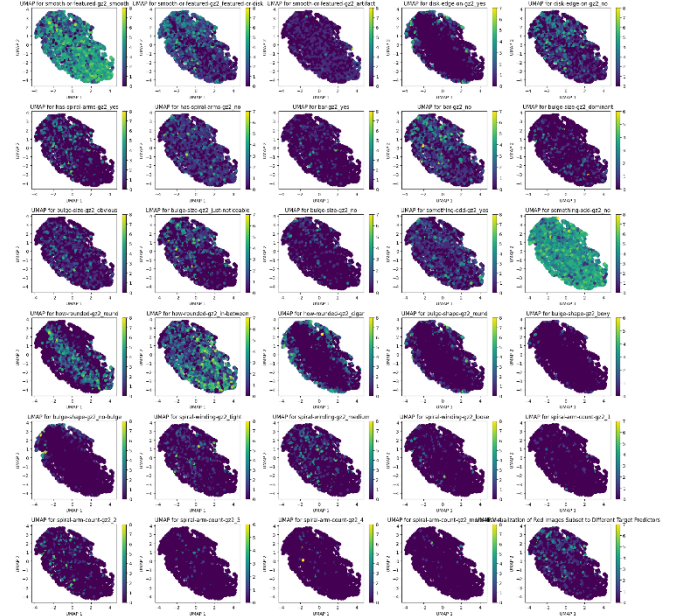


Figure 5: UMAP analysis conducted for each target label

B. Review

The most accurate model was the ensemble learner with 41.39% accuracy on the validation test. The second most successful model was the tuned CNN with 40.15% accuracy on the validation test. However, while the DNN didn't

outperform the other techniques it present a few interesting surprises that may be further explored.

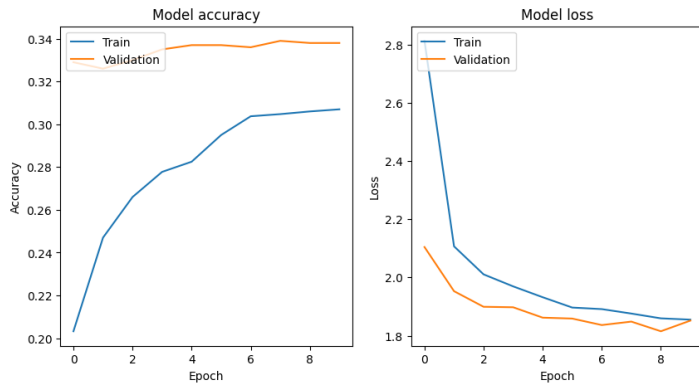


Figure 6: DNN model performance

First, was how well it generalized with relatively low computation cost when compared to other models. While other methods were slow and consumed much of the available computation capital, DNN was quick and unproblematic. Second, while running several more epochs for the other models could have translated in a few more hours of rendering, doubling the epoch for the DNN didn't meaningfully elongate the running time.

X. ADDRESSING SHORTCOMINGS

A reoccurring issue the research encountered was limited computational resources. Training an ensemble learner is quite difficult without dedicated hardware or paying for computation time. Therefore, progress can often be limited and hard to capitalize on. Even though the model's performance ended up being suboptimal, the work improving the performance gave a deeper insight into the general workings of machine learning and helped gain a deeper appreciation towards how truly complicated our universe is.

A. Conclusions

Given the opportunity to restart the project I would have expanded more energy on choosing a solid data set I understood how to work with and manage. So much of the work involved in training machine learning models has to do with having a deep understanding of the data you work with which makes adapting large unfamiliar datasets to something you understand how to deliver to the model a quite challenging task if you must download your data set every time.

Another lesson would be to start small and build up. In general it is easier to construct a successful model and then scale it to a larger data set, however trying to construct a model when computation costs are high is incredibly difficult. Starting from a computationally cheap model and iterating through subtle alteration to the model hyperparameters looking for improvements is easier than restructuring the whole ensemble configuration.

B. Future Work

The field of automating astronomical analysis is in its starting phase. I strongly believe there is a lot more room both for image analysis and image enhancement. To continue to do that we must ensure there is both dataset and other resources so individual researchers can still participate.

To continue this research, several types of ensemble learners can be constructed each capitalizing on different elements of telemetry sources.

Wavelength Stacking: there are a variety of unique mediums in which we use to survey the universe. Since interaction with different objects causes slight variations in the light we capture overlaying a picture of the object in visual spectrum with radio telemetry can help identify previously unknown relationships. Combining learners that survey individual spectrums helps increase the range of data analyzed without running the risk of underfitting.

Wavelength Boosting: Starting at one end of the spectrum of wavelength and boosting towards the other end of the spectrum may reveal hidden relationships of how structures may appear. Through offering perspectives varying from low intensity with plenty of available information and noise to high intensity with little available information and noise in succession, the model may uncover more difficult to identify structures.

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